



Dynamic light scattering in mixed network former glass melts

David L. Sidebottom^{*}, Jay McCown, Josh Kevil, Daniel White, Daniel Olabode

Physics Department, Creighton University, Omaha, NE, USA

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ABSTRACT

Photon correlation spectroscopy was used to investigate the primary viscous relaxation in mixed network former glasses in which polymeric sodium metaphosphate is crosslinked by the addition of a second network forming oxide (GeO_2 , B_2O_3 , Bi_2O_3 , Ga_2O_3 or TeO_2). Analysis of the relaxation provides both the glass transition temperature and fragility as a function of mixing. In each system, a companion nuclear magnetic resonance study exists in the literature which provides valuable insight into the complex bond structure of the oxide network. We demonstrate the glass transition temperature is proportional to the average bridging oxygen bond density while the fragility is not. Instead, a topological correction of the bond density, invoked when rigid super-structural network forming units are present, results in a more satisfactory dependence of fragility on the bond structure of the network.

1. Introduction

Properties of network-forming oxide (NFO) glasses are closely defined by the structure of the network including both the variety of bridging oxygen (BO) connections between the glass forming cations as well as the strength of those connections [1–3]. The classic example are alkali silicate glasses in which the addition of an alkali oxide to SiO_2 drives the formation of *non*-bridging oxygens (NBOs) reducing the number of bridging oxygens per Si from $n = 4$ to some average value $\langle n \rangle$ well < 4 . This *depolymerization* of the oxide network results in a sharp decrease in both the glass transition temperature (T_g) and the chemical durability as well as changes in other relevant properties.

One strategy for fine-tuning properties of a glass is to mix two (or more) common NFOs whose properties bookend a desired value [4–8]. Familiar examples of a mixed network former (MNF) glass are borosilicate glasses (e.g., Pyrex) that exhibit high chemical durability and enhanced resistance to thermal shock [2]. Ideally, the mixing of two NFOs would result in properties consistent with the concentration-weighted average of the two endmembers such that a given property could be anticipated by a line extending between the two. But as numerous studies have revealed [4–6], non-linear trends often result wherein a given property deviates (either positively or negatively) from the simple linear prediction. Sodium borophosphate glasses, for example, exhibit a positive deviation in the ion conductivity with potential benefits for the design of modern solid state batteries [4, 6]. These instances in which a property deviates from its linear mixing

are referred to as the mixed network former *effect* and in the case of sodium borophosphates, Raman and nuclear magnetic resonance (NMR) studies suggest the deviation of ion conductivity is related to changes in the distributions of various network forming units which alter the Coulombic traps that in turn limit ion mobility [4–6,9,10].

In recent years, our group has employed photon correlation spectroscopy (PCS) - a dynamic light scattering technique that can monitor the primary viscous mode (the α -relaxation) in a glass forming melt. As the temporal boundary separating liquid from solid, the α -relaxation is fundamental to the glass transition itself. Our studies have focused exclusively on NFO melts as we have endeavored to close a data gap: the vast majority of studies of the α -relaxation pertain to organic liquids with far fewer studies of covalently-bonded inorganic melts. This is particularly unfortunate since the bond *structure* of NFOs can be chemically manipulated to change the degree of polymerization and should then influence the glass-forming *dynamics*.

Our investigations [11–16] have focused mostly on the fragility and how it depends on changes in the BO bond density (the average number of BO connections per NF cation, $\langle n \rangle$). The fragility,

$$m = \left. \frac{d \log_{10} \langle \tau_\alpha \rangle}{d(T_g/T)} \right|_{T \rightarrow T_g}, \quad (1)$$

is a measure of how rapidly viscosity (or the α -relaxation time) increases with cooling near T_g . A repeated theme in these studies is our proposition that a *canonical* pattern exists wherein the fragility, under certain

^{*} Corresponding author.

E-mail address: sidebottom@creighton.edu (D.L. Sidebottom).

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circumstances, follows a universal curve as a function of the average bond density [11]. Instances in which $m(\langle n \rangle)$ deviates from this pattern appear to be the result of rigid super structural units (SSUs) forming in the network. These SSUs (e.g. the small 3-membered boroxol rings present in B_2O_3) often mimic more stable crystalline analogs and behave as a rigid element within the network that resists internal deformations. From a topological perspective, the network units that compose these SSUs possess *redundant* BO connections which we argue should be excised when assessing the average bond density of the network as a whole. A well-defined procedure has been developed [11,14] to perform these corrections leading to a topologically-corrected average bond density $\langle \phi \rangle$ for which $m(\langle \phi \rangle)$ collapses back onto the canonical curve.

In what is likely to be our last installment, we report here our most recent PCS investigations completed in five different MNF glasses, each based on alloying sodium metaphosphate ($NaPO_3$) with a different NFO (GeO_2 , B_2O_3 , Bi_2O_3 , Ga_2O_3 or TeO_2). These five MNF glasses were chosen specifically because quality NMR and Raman studies have already characterized their molecular-level bond structure. Not only do these structural studies allow the BO bond density ($\langle n \rangle$) to be directly calculated but they also provide critical indirect evidence for SSUs hidden within the structure. We find that glass-forming properties of the melt, namely T_g and m , do exhibit a MNF effect but that both are better interpreted on the basis the average bond density that tracks complex structural changes that result from mixing.

2. Experimental

2.1. Sample preparation

Successful light scattering requires samples free of unwanted debris whose scattering might compete with that arising from density fluctuations inherent in the melt itself. To achieve clean samples, high purity chemicals were used and great care was taken to remove sources of parasitic scattering. A stock of sodium metaphosphate was produced by reacting ammonium dihydrogen phosphate (99.999 %, ThermoScientific) with Na_2CO_3 (99.95 %, ThermoScientific) in a Teflon-coated pot on a hot-plate near 300 °C. The carbonate was first dissolved in water, passed through a 0.45 μm filter, and recrystallized before use to remove unwanted scattering matter. The reaction product was lifted from the pot, coarsely ground and reacted to completion (based on mass analysis) in a small muffle furnace overnight at 350 °C and finally ground to a fine powder. Appropriate amounts of this stock were later fused with the other NFO: GeO_2 (99.9999 %, Alfa Aesar), B_2O_3 (99.999 %, ThermoScientific), Bi_2O_3 (99.9995 %, ThermoScientific), Ga_2O_3 (99.999 %, ThermoScientific) or TeO_2 (99.999 %, ThermoScientific) in a small silica crucible at 800 to 1000 °C for approximately 1 h during which time the melt was repeatedly stirred to obtain a homogenous mixture before casting into a brass mold to form transparent ingots. The use of a silica crucible was necessary owing to a reaction of sodium metaphosphate with platinum crucibles that resulted in gray-tinted ingots with substantial parasitic scattering. Chemical analysis of a handful of B_2O_3 – $NaPO_3$ samples, however, showed no detectable levels of silica present.

For PCS studies, one or more of the cast ingots was inserted into a clean silica ampoule (6 mm ID), remelted and lastly degassed under vacuum (100 milliTor) before being transferred to an awaiting optical furnace where measurements would be collected at fixed temperatures approaching T_g from above. In all the MNF systems studied, a point in the compositional range was reached at which the melt became unstable to slow cooling (due to crystallization or phase separation - both of which generate parasitic light scattering). At these compositional extremes some additional PCS was collected using a slightly modified procedure that avoided cooling from the melt. Instead of remelting an ingot, it was left overnight at a temperature near T_g and allowed to slowly flow until it filled the bottom of the ampoule. Limited PCS

measurements could then be conducted at fixed temperatures above T_g until crystallization or phase separation intervened. In spite of the missing degassing step, no irregularities in either T_g or m were evident between the two approaches.

2.2. Photon correlation spectroscopy

Samples prepared as above were transferred to a pre-heated optical furnace which allowed for incident radiation (532 nm, Coherent Verdi) to be focused into the interior of the melt and the radiation scattered at 90° to be analyzed. The scattered light was imaged onto a 50 μm pinhole located approximately 50 cm in front of a photomultiplier tube configured to perform single photon counting. Following amplification and discrimination, photopulses were input to a commercial correlator that computed the intensity-intensity autocorrelation (ACF),

$$C(t) = \frac{\langle I(t)I(t+t) \rangle}{\langle I(t) \rangle^2} = 1 + A_{COH}|S(q, t)|^2, \quad (2)$$

which in turn is related [17,18] to the liquid's dynamic structure factor, $S(q, t)$, through an instrumental constant $A_{COH} \approx 0.80$.

In our previous studies [19–21] of $NaPO_3$, B_2O_3 and GeO_2 , the measured ACF revealed only a single (angle-independent) relaxation process attributed to the α -relaxation and well described by a stretched exponential of the form:

$$S(q, t) = \exp\left\{ - (t/\tau)^\beta \right\}. \quad (3)$$

The temperature dependence of the decay time closely mimics the viscosity [20] and is unambiguous owing to its wavevector independence. However, in the MNF glasses we observe a secondary relaxation process taking place at much longer decay times. First observed in GeO_2 – $NaPO_3$ melts [12] this decay exhibits a scattering angle dependence consistent with diffusion of a scattering object and has been attributed to the diffusion (possibly constrained diffusion) of the added NF cation (i.e., the Ge atoms). Of the present series of five MNF glasses, this second relaxation for GeO_2 , B_2O_3 and Ga_2O_3 , was reasonably weak compared with the α -relaxation and the two decays could be adequately characterized from an unpolarized spectrum. However, for Bi_2O_3 and TeO_2 , this secondary relaxation (illustrated in Fig. 1 for added TeO_2) tended to overwhelm the ACF making the α -relaxation difficult to resolve. In these last two instances, a Glans-Thompson polarizer was added to the collection optics to block the secondary relaxation so the

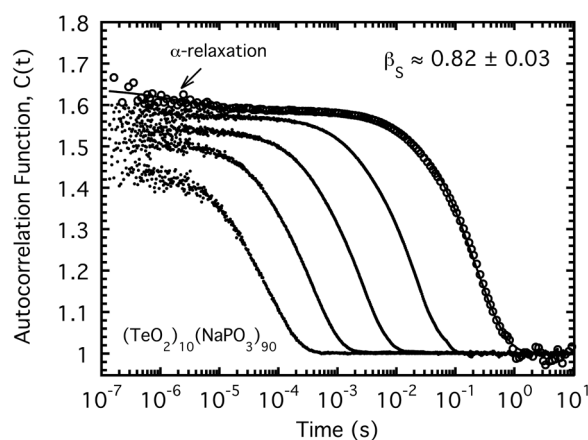


Fig. 1. Autocorrelation function measured in $(TeO_2)_{10}(NaPO_3)_{90}$ glass melt at high temperatures (left to right) of 674.8, 565.8, 482.2, 426.6 and 384.2 °C illustrating the dominance of the slower relaxation with $\beta_s \approx 0.82 \pm 0.03$. The lowest temperature shown includes a two-decay fit (solid line) showing the early emergence of the faster α -relaxation as a small enhancement occurring just below 10^{-5} s.

α -relaxation could be better resolved (see Fig. 2).

3. Analysis

Figs. 1 and 2 illustrate the ACFs observed for the slow (diffusion) and fast (α -relaxation) in $(\text{TeO}_2)_{10}(\text{NaPO}_3)_{90}$, respectively. Each decay was well described by corresponding stretched exponentials:

$$C_F(t) = 1 + A_{\text{COH}} \left| A_F \exp \left[- (t/\tau_F)^{\beta_F} \right] \right|^2, \quad (4)$$

$$C_S(t) = 1 + A_{\text{COH}} \left| A_S \exp \left[- (t/\tau_S)^{\beta_S} \right] \right|^2. \quad (5)$$

Fig. 3 illustrates the temperature dependence of the two relaxation times. At high temperatures only the diffusive mode falls within the time window of the correlator. This relaxation time increases rather gradually with decreasing temperature but eventually disappears out the window of the correlator shortly after the α -relaxation begins to emerge at shorter times. The timescale of the α -relaxation displays a stronger temperature dependence than the slow process as T_g is approached. Here, T_g is conventionally defined as the temperature where the average α -relaxation time reaches 100 s and in Fig. 3, one can see that adding TeO_2 to NaPO_3 results in a progressive increase in the glass transition temperature.

In our previous study [12] of GeO_2 – NaPO_3 melts, we noted that dividing the slow relaxation time by $2.5 (\pm 0.5) \times 10^5$ aligned this secondary process with that of the α -relaxation giving the impression of a continuous process extending from T_g out to very high temperatures and sub-nanosecond timescales. This phenomenon is illustrated for $(\text{TeO}_2)_{10}(\text{NaPO}_3)_{90}$ in Fig. 3 and was similarly observed for all five MNF reported here and deserves some explanation. The shifting may appear empirical but is justified in view of the wavevector dependence of the slow relaxation. The timescale for this slow relaxation is located where it is only because PCS operates at $q = 25 \mu\text{m}^{-1}$ and monitors fluctuations arising from diffusion of the Te cations on a rather large $\Delta r = 2\pi/q \sim 0.25 \mu\text{m}$ displacement. For any alternative probe of dynamics operating at a larger wavevector, the same relaxation would be located at a shorter time. Given the $\tau_S \sim q^{-2}$ dependence [17], our shift by $2.5 (\pm 0.5) \times 10^5$ reveals the diffusion process as it appears near $q \sim 1.2 (\pm 0.1) \text{ \AA}^{-1}$ corresponding to localized motions of the Te cation on the order of $\Delta r \sim 5 \text{ \AA}$. We contend that this is diffusion at its origin; the random kicking that ultimately drives Brownian motion. Here the source for this random kicking is localized motions of the phosphate chains that constitute the α -relaxation itself [12].

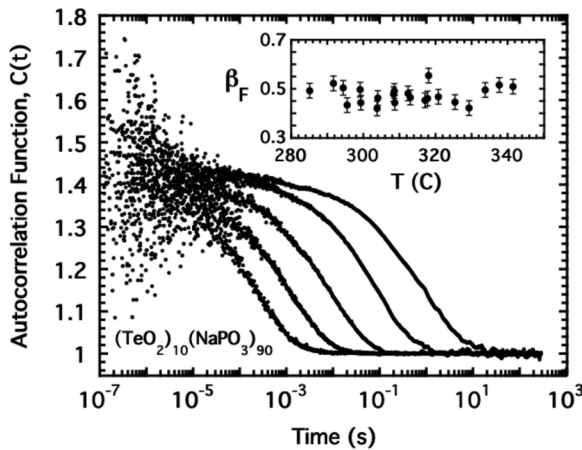


Fig. 2. Autocorrelation functions obtained from depolarized scattering from $(\text{TeO}_2)_{10}(\text{NaPO}_3)_{90}$ at temperatures (left to right) of 333.8, 325.4, 317.1, 308.6 and 299.3 °C. Inset shows the corresponding stretching exponent for this α -relaxation as a function of temperature.

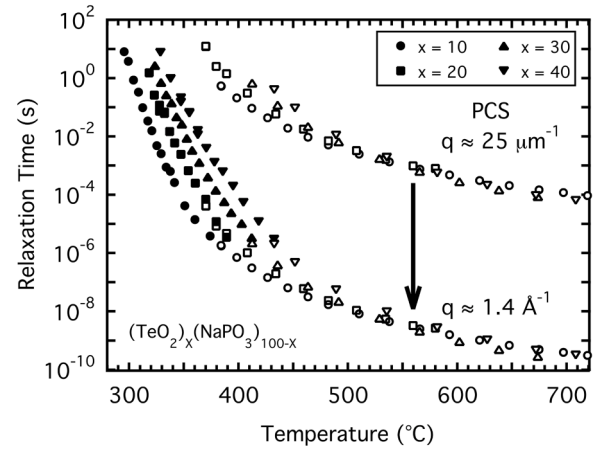


Fig. 3. The fast (solid symbols) and slow (open symbols) relaxation times (defined in Eqs. (4) and (5)) plotted as a function of temperature for $(\text{TeO}_2)_x(\text{NaPO}_3)_{100-x}$ glass melts with $x = 10$ (circles), 20 (squares), 30 (triangles) and 40 (inverted triangles). The slow relaxation times are also shown as they would appear when shifted (as described in the text) to an alternative wavevector range where they appear to align with the (q -independent) α -relaxation.

Fig. 4 illustrates our methods for determining both the glass transition temperature and the fragility index. Because the α -relaxation is non-exponential, it is customary to work with the average relaxation time $\langle \tau_a \rangle = \frac{\Gamma(1/\beta_F)\tau_F}{\beta_F}$, where $\Gamma(z)$ is the gamma function [18]. The glass transition temperature is then obtained by extrapolation of $\langle \tau_a \rangle$ to 100 s. This extrapolation (see Fig. 4) is typically no more than a couple decades from the last measured values. Once the glass transition temperature is determined the abscissa of the plot can be scaled to T_g (see inset to Fig. 4) such that the slope of $\langle \tau_a \rangle$ near $T_g/T = 1$ provides the fragility index as is defined in Eq. (1).

4. Results

In the following, we summarize findings for the behavior of the α -relaxation in each of the five MNF systems studied. The five are arranged in the chronological order in which they were investigated, but also largely in order of increasing coordination of the added NFO (Ge, B enter as 4-coordinated, while Bi and Ga are mostly 6-coordinated). In

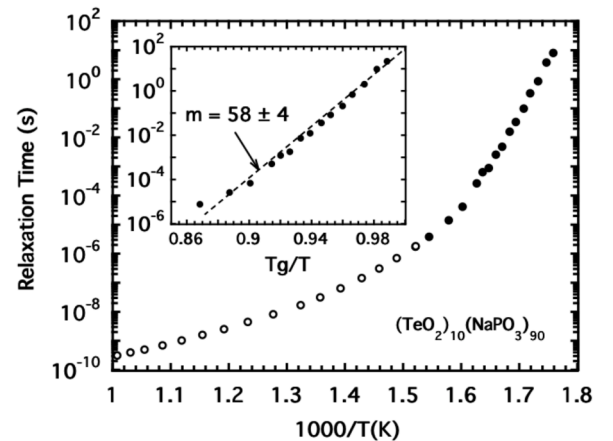


Fig. 4. The α -relaxation time of $(\text{TeO}_2)_{10}(\text{NaPO}_3)_{90}$ (including data from the slow relaxation shifted as described in the text) plotted against inverse temperature. The glass transition temperature is evaluated by the short extrapolation to 100 s. Inset is the same in a scaled T_g representation from which the fragility index can be determined from the slope.

every series, we find that adding the NFO to sodium metaphosphate ($\langle n \rangle = 2$) causes crosslinking of the phosphate chains which systematically increase the average bond density and so too the glass transition temperature. However, while the fragility mostly decreases with the initial addition of the NFO, $m(\langle n \rangle)$ displays no common pattern among the five MNF systems and consistently fails to track the canonical curve which was previously identified. This canonical curve developed from an early comparison of the fragilities measured in alkali phosphates [19] and in a series of chalcogenides [22], $\text{As}_x\text{Ge}_{100-2x}$. The average bond density in the alkali phosphates is given by $\langle n \rangle$ because the network flexes about these BO connections, while the bond density of the chalcogenide is provided by $\langle r \rangle$ the average number of covalent bonds per atom.

Remarkably, as seen in the figures that follow, both $m(\langle n \rangle)$ for the phosphates and $m(\langle r \rangle)$ for the chalcogenide coincide over the entire range from $\langle r, n \rangle = 2$ to 3 as m decreases with increasing density of connections. This coincidence is taken to imply that the two networks assume a similar topology when coarse graining is applied to properly identify the rigid objects (tetrahedra or atoms, respectively) interconnected to form a flexible structure. We refer to their common curve as "canonical" because there are numerous instances in NFOs for which $m(\langle n \rangle)$ radically deviates from this curve. For example, alkali borates undergo a conversion [23] from 3-coordinated to 4-coordinated boron when alkali oxide is added that increases $\langle n \rangle$ but surprisingly increases the fragility too [16]. As it happens the network structure of alkali borates is riddled with small, superstructural units (SSUs) such as 3-membered boroxol rings and 4-membered diborate units [24]. We believe it is the presence of these SSUs that cause the deviation of $m(\langle n \rangle)$ from our canonical curve as both the phosphate and chalcogenide represent networks presumably free of SSUs. In an effort to interpret such deviations, we proposed a topological correction be applied to rigid SSUs to excise redundant BO connections they contain and so reassess the effective bond density, $\langle \phi \rangle$. In many cases [11,14,25] we have shown that, with reasonable assumptions, this topological correction results in $m(\langle \phi \rangle)$ collapsing onto the canonical curve.

The primary objective of this current study is to explore whether the MNF effects that are commonly observed only in glass solids might be reflected in the liquid state as well. The lesser objective is to employ details gleaned from the NMR studies (including P speciation, heteroatomic and homoatomic bond distributions, and coordination changes in the added NFO) to speculate on possible SSUs that might be present in these systems. In every instance, we can, with reasonable assumptions, invoke topological corrections in the bond connectivity arising from these SSU that result in $m(\langle \phi \rangle)$ agreeing with the canonical curve.

4.1. GeO_2

Samples of $(\text{GeO}_2)_x(\text{NaPO}_3)_{100-x}$ were studied [12] by PCS out to a maximum of $x = 40$ mol % GeO_2 before the melts became unstable due to phase separation (with strong opalescence). Results are presented in Table 1 as well as in Fig. 5. Across this compositional range we find that the α -relaxation is well-described by a stretched exponential with a

Table 1

Results for PCS in GeO_2 – NaPO_3 glass melts and bond density analysis. Typical uncertainties for T_g (± 10 °C), fragility (± 8 %) and non-exponentiality parameter (± 0.03).

GeO_2 (mol %)	T_g (°C)	β_{FAST}	β_{SLOW}	Fragility	$\langle n \rangle$	$\langle \phi \rangle$
5	314	0.37	0.84	56	2.10	2.10
10	367	0.44	0.78	51	2.19	2.14
15	396	0.41	0.71	44	2.31	2.20
20	458	0.41	0.64	48	2.39	2.19
25	441	0.46	0.59	55	2.49	2.13
30	479	0.44	0.58	60	2.58	2.12
35	483	0.45	0.56	56	2.69	2.13
40	487	0.48	0.58	60	2.79	2.15

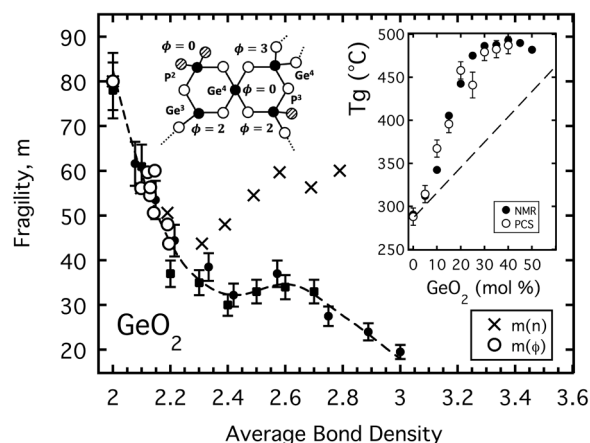


Fig. 5. The fragility index plotted against average bond density. Solid symbols with dashed line indicate the canonical curve based on alkali phosphates (circles) and chalcogenides (squares) discussed in the text. Crosses refer to $(\text{GeO}_2)_x(\text{NaPO}_3)_{100-x}$ glass melts as a function of $\langle n \rangle$ while open circles are as a function of $\langle \phi \rangle$. Inset shows the glass transition temperatures from PCS (open circles) and literature [26] (closed circles) and dashed line for linear mixing. The cartoon illustrates the topological corrections applied to potential ring structures discussed in the text.

reasonably composition independent $\beta_F \approx 0.44 (\pm 0.04)$. The glass transition temperatures, determined from extrapolation of $\langle \tau_\alpha \rangle$ to 100 s, show a steady increase with added GeO_2 - increasing rapidly at low x and less so at $x > 25$ mol %. As shown in the inset to Fig. 5, these T_g values are in excellent agreement with those reported by DSC techniques [26].

In Table 1, the fragility index initially decreases with added GeO_2 but then begins increasing above $x = 15$ mol %. To test how $m(\langle n \rangle)$ behaves, we extract from the NMR study by Ren and Eckert [26] details on the bonding configurations of both the P and Ge atoms. The NMR study reports that Ge enters the network structure as either Ge^3 or Ge^4 having $n = 3$ or 4 BO bonds, respectively, and that the P appear as either P^2 or P^3 having $n = 2$ or 3 BO bonds, respectively. Using the species concentrations provided [26], the average BO bond density was calculated as:

$$\langle n \rangle = f_P(2f_{P^2} + 3f_{P^3}) + f_{Ge}(3f_{Ge^3} + 4f_{Ge^4}), \quad (6)$$

where the fractions $f_P = (100 - x)/100$ and $f_{Ge} = x/100$ are the total fraction of cations M ($= P$ or Ge) given by the chemical formula and f_{Mn} is the sub-fraction that have n BO connections. Values of $\langle n \rangle$ so calculated are included in Table 1 and are used to plot $m(\langle n \rangle)$ in Fig. 5. With the initial addition of GeO_2 , $m(\langle n \rangle)$ at first tracks along with the canonical curve but then begins to deviate as $m(\langle n \rangle)$ begins increasing. The goodness of fit [27] between $m(\langle n \rangle)$ and the canonical curve (taken to be represented by the dashed line and assuming an error of 10 % in m) is $\chi^2_{\text{red}} = 11.7$ and expectedly poor.

In addressing this deviation, we reflected on an earlier analysis of binary alkali germanates where the presence of small three-membered rings required topological corrections. We proposed [12] that such small rings might still be present in this MNF, but could randomly involve both P and Ge units. The cartoon inset to Fig. 5 illustrates many of the possible configurations for Ge^3 , Ge^4 , P^2 and P^3 units that participate in a ring. Generally, the presence of any redundant BO bond leads to a reduction of the connectivity of the unit. For example, both the Ge^3 and P^3 units in a ring topologically function to connect the ring to the external network and so one of the two BO bonds in the ring is redundant. Correcting for this results in a revised connectivity of $\phi = 2$ for either of these units. In the rare instances that a Ge^4 unit happens to participate in two rings, simultaneously, the unit is assigned zero connectivity [15] as it makes no external connections to the remaining

network. In a similar fashion, a P^2 unit participating in a ring functions as a "dangling" bond [28] offering $\phi = 0$ connectivity. By assuming then that some fraction, $g_R(x)$, of the Ge^3 , Ge^4 , P^2 and P^3 units participate in a ring (and $g_{NR}(x) = 1 - g_R(x)$, do not) we obtained:

$$\langle \phi \rangle = g_{NR} [f_P(2f_{P2} + 3f_{P3}) + f_{Ge}(3f_{Ge3} + 4f_{Ge4})] + g_R [g_{NR}(2f_{P2} + 3f_{P3}) + f_{Ge}(2f_{Ge3} + 3f_{Ge4})] + g_R(0 \times f_{Ge4}). \quad (7)$$

In the above equation, the first term represents normal units (not participating in rings) while the second term represents units in rings which are further divided into those in a single ring and the small fraction of Ge^4 which could be present in two rings simultaneously. This proposed random distribution of P and Ge units is also supported by the NMR study which reported nearly random bonding with only a modest preference for heteroatomic (P - O - Ge) connections. The functional form for $g_R(x)$ was based on changes in the strength of a Raman mode attributed to 3-membered rings. Details can be found elsewhere [12], but the resulting $m(\langle \phi \rangle)$ that is included in Fig. 5 shows improved agreement with the canonical mastercurve and an improved goodness of fit ($\chi^2_{red} = 0.3$).

4.2. B_2O_3

Samples of $(B_2O_3)_x(NaPO_3)_{100-x}$ glass melts were investigated by PCS up to $x = 18$ mol % B_2O_3 beyond which they became unstable to phase separation/crystallization. Results of the PCS analysis for the α -relaxation are presented in Table 2 and illustrated in Fig. 6. Across this compositional range we find that the α -relaxation is well-described by a stretched exponential with β_F increasing slightly from 0.42 to 0.52 with increasing B_2O_3 . The glass transition temperature increases with added B_2O_3 and is consistent with B entering the network as a crosslinking unit. Our values of T_g obtained from analysis of the PCS agree well with literature up to $x = 10$ mol % but deviate at higher concentrations.

According to the NMR study [10], all boron atoms added to the network appear exclusively as B^4 with $n = 4$ BO connections and were further subdivided according to the number ($q = 3$ or 4) of the four BO connections that are made to P atoms, B^4_{qp} . The NMR suggests that as x increases from $x = 0$ to $x = 20$ mol %, the average value of q decreases and tends toward a value of $q = 3$ which is partly expected as B - O - B connections naturally become more favorable at higher loading. As for the P speciation, there are mostly only three P^n_{qB} species aside from a small fraction of P^4_{4B} and these are provided in Table 3 (as reported in the NMR study). Taken together with the total fractions of B ($f_B = 2x/(100 + x)$) and P ($f_P = (100 - x)/(100 + x)$) atoms present, the average BO bond density could be calculated as:

$$\langle n \rangle = f_P(2f_{P2} + 3f_{P3} + 4f_{P4}) + 4f_B, \quad (8)$$

and is provided in Table 2. As can be seen in Fig. 6, the $m(\langle n \rangle)$ again deviates from the canonical curve but, unlike adding GeO_2 , decreases steadily and mimics the curve albeit vertically displaced. The goodness of fit, $\chi^2_{red} = 9.6$, again suggests poor agreement.

We suppose this deviation to be a result of unidentified SSUs present

Table 2

Results for PCS in B_2O_3 - $NaPO_3$ glass melts and bond density analysis. Typical uncertainties for T_g (± 10 °C), fragility (± 8 %) and non-exponentiality parameter (± 0.03).

B_2O_3 (mol %)	T_g (°C)	β_{FAST}	β_{SLOW}	Fragility	$\langle n \rangle$	$\langle \phi \rangle$
4	308	0.42	0.45	59	2.24	2.10
6	306	0.43	0.61	50	2.36	2.17
10	329	0.46	0.57	38	2.57	2.23
12	362	0.47	0.59	47	2.67	2.24
14	375	0.50	0.56	41	2.77	2.26
18	395	0.52	0.40	35	2.96	2.27

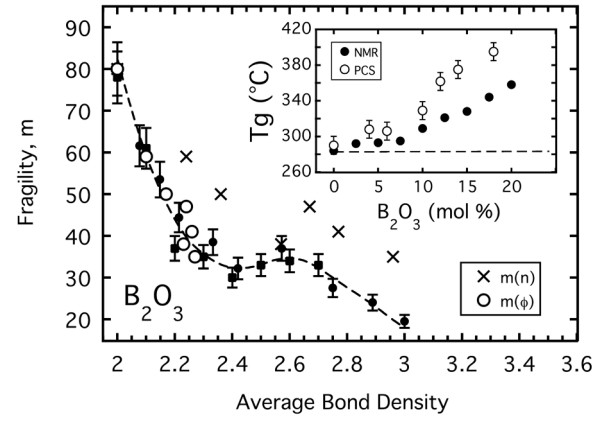


Fig. 6. The fragility index plotted against average bond density. Solid symbols with dashed line indicate the canonical curve based on alkali phosphates (circles) and chalcogenides (squares) discussed in the text. Crosses refer to $(B_2O_3)_x(NaPO_3)_{100-x}$ glass melts as a function of $\langle n \rangle$ while open circles are as a function of $\langle \phi \rangle$. Inset shows the glass transition temperatures from PCS (open circles) and literature [10] (closed circles) and dashed line for linear mixing.

Table 3

Phosphor speciation as reported by NMR for B_2O_3 - $NaPO_3$ glasses [10].

B_2O_3 (mol %)	P^2_{0B} (%)	P^2_{1B} (%)	P^3_{2B} (%)	P^4_{4B} (%)
2.5	84.6	10.2	5.1	0
5	74.7	14.7	7.3	3.1
7.5	57.8	25.9	13.0	3.2
10	46.4	31.3	19.0	3.2
12.5	36.2	35.2	26.2	2.4
15	28.7	36.0	32.1	3.2
17.5	19.1	38.4	38.3	4.1
20	11.1	38.9	44.3	5.6

and we examine in greater detail the NMR speciation listed in Table 3. Firstly, it is worth noting that the authors of the NMR study [10] of $(B_2O_3)_x(NaPO_3)_{100-x}$ glasses emphasized that "the connectivity distribution is highly nonstatistical" and the strange absence of many possible P^n_{qB} units suggest that P^2_{1B} and P^3_{2B} "have special stability in this glass system". The authors also emphasize that heteroatomic bonding of the form B - O - P is dominant; that is, greater than expected from a purely random model of network connections. Secondly, in Table 3 we detect a pattern in the P speciation in which the P^2_{0B} units exclusive to $NaPO_3$ are being replaced by only P^2_{1B} and P^3_{2B} in *strict ratios* that depend upon the compositional extreme. At low $x < 10$ mol %, we see that the conversion results in exactly *twice* as many P^2_{1B} units as P^3_{2B} , while at higher $x \geq 15$ mol %, the conversion favors a nearly equivalent division into P^2_{1B} and P^3_{2B} units. Taken together with the NMR observation that B^4_{qp} units evolve toward $q = 3$ at higher concentrations, we propose the two potential SSUs that are illustrated in Fig. 7. The first of these are rings (the smallest with 4 BOs is shown but rings with 4, 6, 8 etc. could also work) that are constructed of alternating B^4_{4P} and P^3_{2B} units. This ring structure would satisfy the 2:1 ratio of $P^2_{1B}:P^3_{2B}$ observed at low x and has $q = 4$ owing to the absence of any B - O - B connection. The second possible structure is a "diborate"-like unit in which a single B - O - B link is introduced. This structure would satisfy the 1:1 ratio of $P^2_{1B}:P^3_{2B}$ observed at higher x and is consistent with the transition toward $q = 3$ reported in the NMR study. Additional support for the diborate structure is found in a study of sodium borophosphate glasses by another group [29], who proposed this same diborate-like super-structural unit to model sodium ion diffusion.

The procedure for topologically correcting redundant BO bonds in a diborate unit was introduced [16] previously in the analysis of binary alkali borate glasses. Following that same protocol, each P^3_{2B} unit in

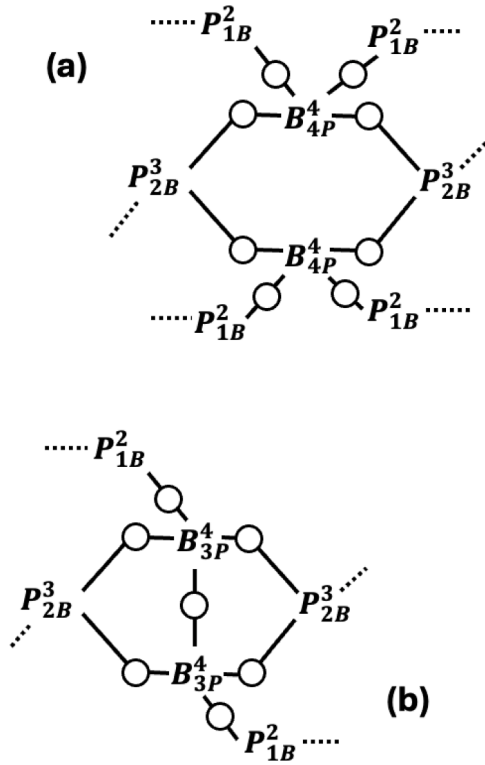


Fig. 7. Possible super structural units consistent with connectivity distributions reported for $(\text{Bi}_2\text{O}_3)_x(\text{NaPO}_3)_{100-x}$ by NMR [10]. (a) A 4-membered B and P ring unit favored at low x . (b) A diborate-like unit favored at high x .

Fig. 7b assumes $\phi = 2$ (owing to a single redundant BO bond) while each B_{3P}^4 units assumes $\phi = 2$ (owing to two redundant BO bonds). However, as for the ring structures (Fig. 7a) that are likely present at lower x , we do not presently believe such large rings require topological correction. Indeed, large rings with many repeating units (>4 BO) are generally not seen as being "rigid" elements due to their ability to be deformed internally (i.e., "crumpled") and it is unclear at what decreasing size the "rigidity" of a ring actually develops. Currently, we maintain the opinion that a 4-membered ring like that illustrated in Fig. 7a (or any larger such ring) still contains sufficient internal flexibility to refrain from treating it as a SSU that would require topological corrections.

Assuming then that only those P_{2B}^3 and B_{3P}^4 units that occur in diborate units require correction, we can calculate the revised bond density as:

$$\langle \phi \rangle = f_B(2g_{DI} + 4(1 - g_{DI})) + 2f_B g_{DI} + 3(f_P f_{P3} - f_B g_{DI}) + f_P(2f_{P2} + 4f_{P4}) \quad (9)$$

where $g_{DI}(x)$ is the fraction of B^4 units that participate in diborate structures. In the expression, the first term accounts for all the B^4 units (those $f_B g_{DI}$ in diborate units with $\phi = 2$ and the remainder with $n = 4$). The second term accounts for the subset of P^3 units in diborate units which must equal the number of B^4 units in diborate units. The third term accounts for all the remaining P^3 units and the last term represents P^2 and P^4 units with normal connectivity. All that is missing is a model for the compositional dependence of $g_{DI}(x)$. To minimize the degree of freedom in the problem and to capture the trend of q approaching $q = 3$ at the compositional extreme, we have opted to test a linear model of the form $g_{DI}(x) = A + Bx$. In Fig. 6 (and in Table 2) we report the results obtained for $A = 50\%$ and $B = 3/2$. These parameters produced $m(\langle \phi \rangle)$ which more closely matches the canonical curve ($\chi_{red}^2 = 0.8$) and resulted in q ranging from 3.5 to 3.2 which is consistent with values reported in the NMR study [10].

4.3. Bi_2O_3

Our PCS study of $(\text{Bi}_2\text{O}_3)_x(\text{NaPO}_3)_{100-x}$ glass melts [30] extended only to $x = 10$ mol % beyond which samples became unstable to phase separation and crystallization when cooled slowly. Results of the PCS study are summarized in Table 4. Similar with GeO_2 discussed earlier, the α -relaxation is well described by a stretched exponential with $\beta_F \approx 0.43 (\pm 0.03)$ with no discernable compositional dependence. The fragility index decreases and the glass transition temperature increases steadily with the addition of Bi_2O_3 . Values of T_g attained from PCS analysis agree well with others [31] as shown in the inset to Fig. 8.

In the corresponding NMR study [31], the authors reported that Bi appears only in an octahedral configuration (Bi^6) while the P species are limited to P^2 and P^3 units. From the P species distributions provided by NMR, the average BO bond density was calculated as:

$$\langle n \rangle = f_P(2f_{P2} + 3f_{P3}) + 6f_{Bi}, \quad (10)$$

and is included in Table 4. Owing to the 6-fold coordination of the Bi atoms, $\langle n \rangle$ increases rapidly with x but $m(\langle n \rangle)$ again deviates from the canonical curve ($\chi_{red}^2 = 27$).

A unique feature, stressed by the authors of the NMR study, is how the Bi octahedral make strictly edge-sharing connections with P units (like the one illustrated in Fig. 8). These edge-sharing connections lower the topological connectivity of every Bi atom from $n = 6$ to $\phi = 3$ and, as it happens [31], limits the connectivity of all the P units to $\phi = 2$. In this instance the topologically corrected bond density is given simply by:

$$\langle \phi \rangle = 2f_P + 3f_{Bi}, \quad (11)$$

and as shown in Fig. 8 provides good agreement of $m(\langle \phi \rangle)$ with the canonical curve ($\chi_{red}^2 = 0.8$).

4.4. Ga_2O_3

Samples of $(\text{Ga}_2\text{O}_3)_x(\text{NaPO}_3)_{100-x}$ glass melts were investigated by PCS up to $x = 20$ mol % Ga_2O_3 beyond which they became unstable to phase separation/crystallization. Results of the PCS analysis for the α -relaxation are presented in Table 5 and illustrated in Fig. 9. In good agreement with others [32], the T_g determined from analysis of PCS increases with increasing Ga_2O_3 up to about 15 mol % and then plateaus before increasing again at $x > 20$ mol % (see inset to Fig. 9).

The corresponding NMR study [32] of these same MNF glasses reported that Ga is predominantly $n = 6$ at low concentrations ($x < 15$ mol %) but transitions to $n = 4$ at higher concentrations of Ga_2O_3 . The fractions of Ga in Ga^6 and Ga^4 reported in the NMR study are reproduced in Table 6 along with those major P species obtained from the deconvolution of the ^{31}P MAS spectra [32]. Using these P and Ga distributions, we can calculate the average BO bond density as:

$$\langle n \rangle = f_P(2f_{P2} + 3f_{P3} + 4f_{P4}) + f_{Ga}(6f_{Ga6} + 4f_{Ga4}), \quad (12)$$

and this is included in Table 5. The resulting $m(\langle n \rangle)$ plotted in Fig. 9 misses the canonical curve in a manner similar to Bi_2O_3 (see Fig. 8) with a similarly poor goodness of fit ($\chi_{red}^2 = 26$).

Of the five MNF glass systems reported in this present study, the

Table 4

Results for PCS in Bi_2O_3 – NaPO_3 glass melts and bond density analysis. Typical uncertainties for T_g (± 10 °C), fragility (± 8 %) and non-exponentiality parameter (± 0.03).

Bi_2O_3 (mol %)	T_g (°C)	β_{FAST}	β_{SLOW}	Fragility	$\langle n \rangle$	$\langle \phi \rangle$
2	282	0.40	0.58	65	2.28	2.04
4	293	0.42	0.58	58	2.54	2.08
7	301	0.45	–	50	2.90	2.13
10	313	0.44	–	50	3.23	2.18

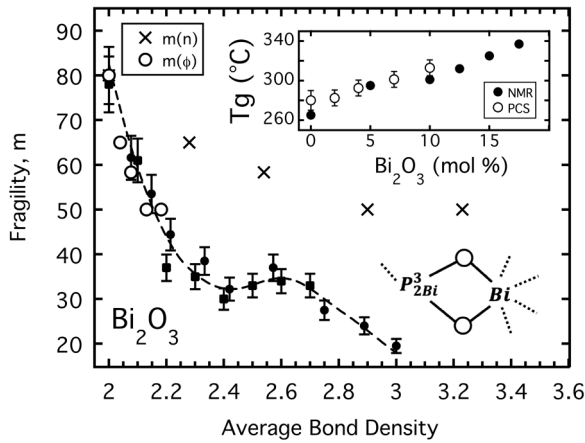


Fig. 8. The fragility index plotted against average bond density. Solid symbols with dashed line indicate the canonical curve based on alkali phosphates (circles) and chalcogenides (squares) discussed in the text. Crosses refer to $(\text{Bi}_2\text{O}_3)_x(\text{NaPO}_3)_{100-x}$ glass melts as a function of $\langle n \rangle$ while open circles are as a function of $\langle \phi \rangle$. Inset shows the glass transition temperatures from PCS (open circles) and literature [31] (closed circles). The cartoon illustrates the topological implications of potential edge-sharing connections.

Table 5

Results for PCS in $\text{Ga}_2\text{O}_3\text{--NaPO}_3$ glass melts and bond density analysis. Typical uncertainties for T_g (± 10 °C), fragility (± 8 %) and non-exponentiality parameter (± 0.03).

Ga ₂ O ₃ (mol %)	T_g (°C)	β_{FAST}	β_{SLOW}	Fragility	$\langle n \rangle$	$\langle \phi \rangle$
1	290	0.41	0.55	74	2.11	2.02
2	306	0.39	0.66	64	2.22	2.03
3	303	0.33	0.66	61	2.33	2.05
5	323	0.40	0.66	55	2.58	2.08
10	361	0.44	0.56	48	3.11	2.14
15	401	0.46	–	54	3.30	2.16
20	384	0.56	–	50	3.29	2.13

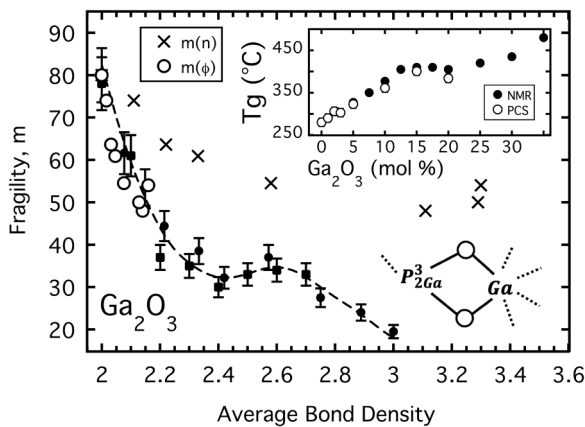


Fig. 9. The fragility index plotted against average bond density. Solid symbols with dashed line indicate the canonical curve based on alkali phosphates (circles) and chalcogenides (squares) discussed in the text. Crosses refer to $(\text{Ga}_2\text{O}_3)_x(\text{NaPO}_3)_{100-x}$ glass melts as a function of $\langle n \rangle$ while open circles are as a function of $\langle \phi \rangle$. Inset shows the glass transition temperatures from PCS (open circles) and literature [32] (closed circles). The cartoon illustrates the topological implications of potential edge-sharing connections.

Ga_2O_3 alloy has posed the greatest challenge for identifying potential SSUs that might account for the deviation of $m(\langle \phi \rangle)$ from the canonical curve. Unlike the B_2O_3 case above, the P distributions in Table 6 do not appear to suggest any special ratios nor other non-statistical trends.

Table 6

Phosphor and Gallium speciation as reported by NMR for $\text{Ga}_2\text{O}_3\text{--NaPO}_3$ glasses [32]. Other P_{Ga}^n units with <2 % across the entire compositional range are omitted.

Ga ₂ O ₃ (mol %)	$P_{0\text{Ga}}^2$ (%)	$P_{1\text{Ga}}^2$ (%)	$P_{1\text{Ga}}^3$ (%)	$P_{2\text{Ga}}^3$ (%)	$P_{3\text{Ga}}^3$ (%)	Ga^{VI} (%)	Ga^{IV} (%)
2.5	73.5	11.9	8.2	4	0	78.5	21.5
5	51.7	18.3	15.1	11.4	0	80.0	20.0
7.5	29.5	29.1	20.1	17.3	0	71.6	28.4
10	7.6	34.3	24.6	26.1	0	77.2	22.8
12.5	0	36.3	18.5	31.9	5.9	66.7	33.3
15	0	33.3	11.6	32.8	16.3	61.1	38.9
17.5	0	28.9	6.9	27.8	27.4	45.2	54.8
20	0	24.9	4.7	21.6	36.6	38.4	61.6
22	0	16	2.8	13.5	52	36.2	63.8
25	0	3.8	0.8	6.2	70.5	28.8	71.2

Undeterred, we consider one possible (*ad hoc*) model suggested only by the similarity of $m(\langle n \rangle)$ to that observed in Bi_2O_3 . We test the hypothesis that (like Bi_2O_3) all the Ga^6 and Ga^4 units in this glass system form only edge-sharing units in which case Eq. (11) would be slightly reformulated as:

$$\langle \phi \rangle = 2f_P + f_{\text{Ga}}(3f_{\text{Ga}^6} + 2f_{\text{Ga}^4}), \quad (13)$$

to incorporate the additional presence of edge-sharing Ga^4 units. In Fig. 9 we see that this simple model is largely successful in achieving a match of $m(\langle \phi \rangle)$ to the canonical curve ($\chi_{\text{red}}^2 = 1.4$).

4.5. TeO_2

PCS studies were conducted in $(\text{TeO}_2)_x(\text{NaPO}_3)_{100-x}$ glass melts up to $x = 60$ mol % TeO_2 before instability to crystallization developed. Results of the analysis are presented in Table 7. The α -relaxation was well described by a stretched exponential whose β_F increased slightly with x from $\beta_F = 0.42$ (± 0.03) to 0.52 (± 0.03). As shown in the inset to Fig. 10, the glass transition temperatures determined from PCS are in fair agreement with that reported by others [33] with the possible exception of the samples at $x = 6$ and 60 mol % which appear to be possible outliers.

In the corresponding NMR investigation of these same glasses [33], the authors emphasized a "nonstatistical connectivity distribution" in which the formation of homoatomic (P - O - P and Te - O - Te) connections are favored over heteroatomic (P - O - Te) connections as is indicated in Table 8. The mixing of the two NF glasses produces no new structural units and no sharing of the Na_2O modifier. Consequently, all the P species retain two non-bridging oxygen (charge compensated by a Na ion) and make only $n = 2$ BO connections throughout the entire compositional range with the $P_{0\text{Te}}^2$, $P_{1\text{Te}}^2$, or $P_{2\text{Te}}^2$ fractions as tabulated in Table 8. Since the Te atoms appear as distorted tetrahedra (antiprismatic units) with $n = 4$ BO bonds, the average BO bond density can be calculated simply as:

Table 7

Results for PCS in $\text{TeO}_2\text{--NaPO}_3$ glass melts and bond density analysis. Typical uncertainties for T_g (± 10 °C), fragility (± 8 %) and non-exponentiality parameter (± 0.03).

TeO ₂ (mol %)	T_g (°C)	β_{FAST}	β_{SLOW}	Fragility	$\langle n \rangle$	$\langle \phi \rangle$
2	284	0.40	0.82	63	2.04	2.04
4	269	0.44	0.82	64	2.08	2.08
6	262	0.46	0.82	60	2.12	2.12
10	280	0.48	0.79	58	2.20	2.16
20	296	0.49	0.79	50	2.40	2.12
30	308	0.51	0.84	52	2.60	2.13
40	313	0.53	0.86	44	2.80	2.16
50	315	0.53	–	47	3.00	2.20
60	296	0.51	–	49	3.20	2.24

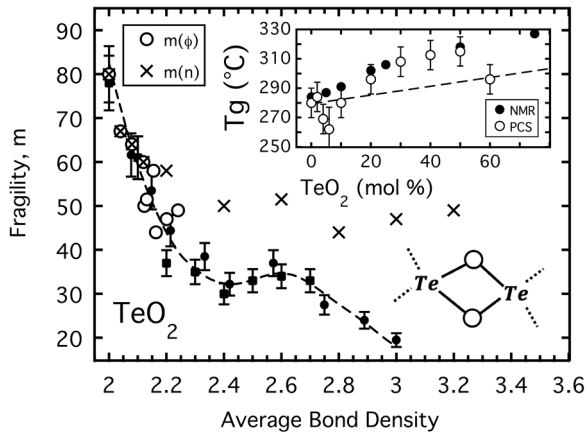


Fig. 10. The fragility index plotted against average bond density. Solid symbols with dashed line indicate the canonical curve based on alkali phosphates (circles) and chalcogenides (squares) discussed in the text. Crosses refer to $(\text{TeO}_2)_x(\text{NaPO}_3)_{100-x}$ glass melts as a function of $\langle n \rangle$ while open circles are as a function of $\langle \phi \rangle$. Inset shows the glass transition temperatures from PCS (open circles) and literature [33] (closed circles) and dashed line for linear mixing. The cartoon illustrates the topological implications of potential edge-sharing connections.

Table 8

Phosphor speciation and hetero- versus homoatomic bonding as reported by NMR for TeO_2 - NaPO_3 glasses [33]. Values in parenthesis are the deviation from a model of random bonding (for deviations greater than 4 %). All values are rounded to the nearest whole percent.

TeO_2 (mol %)	$P_{2\text{Te}}^2$ (%)	$P_{2\text{Te}}^2$ (%)	Te - O - Te (%)	P - O - Te (%)
5	7 (-10)	3 (-)	0 (-)	17 (-)
10	14 (-16)	7 (-)	0 (-)	23 (-6)
15	22 (-17)	10 (-)	6 (-)	30 (-8)
20	33 (-11)	9 (-)	21 (+10)	33 (-12)
25	44 (-)	5 (-11)	31 (+15)	32 (-16)
50	76 (+32)	14 (-30)	50 (+6)	34 (-10)
75	51 (+27)	4 (-70)	77 (-)	19 (-5)

$$\langle n \rangle = 2f_p + 4f_{\text{Te}}, \quad (14)$$

and is tabulated in Table 7. As shown in Fig. 10, $m(\langle n \rangle)$ is seen to initially match the canonical curve up until $x \approx 15$ mol % TeO_2 but then begins to deviate and approach a nearly constant fragility of approximately $m \approx 50$. The overall goodness of fit is $\chi_{\text{red}}^2 = 13.9$.

We believe that the nonstatistical connectivity distribution reported in the NMR investigation provides clues to a possible SSU. We propose that this SSU arises from edge-sharing connections formed between two Te atoms (forming a $\text{Te-O}_2\text{-Te}$ unit similar to that shown by a cartoon inset to Fig. 10). Support for this motif arises from multiple sources. Firstly, the early X-ray diffractions studies of TeO_2 glasses by Brady [34] concluded that not all Te units could be connected by only corner sharing and that, in violation of Zachariasen's rules [35], some fraction would necessitate edge-sharing connections. Secondly, in recent Raman investigations [36] of TeO_2 , a dominant mode near 655 cm^{-1} wavenumber was assigned to the in-phase symmetric "breathing" of $\text{Te-O}_2\text{-Te}$ units. We observe in the Raman spectra of $(\text{TeO}_2)_x(\text{NaPO}_3)_{100-x}$ glasses [33] a mode of similar energy present at 615 cm^{-1} that first appears near $x = 10$ % and increases in intensity between $x = 10$ and $x = 25$ mol %. Given its close proximity to the $\text{Te-O}_2\text{-Te}$ breathing mode seen in pure TeO_2 , we speculate that this mode results from the development of edge-sharing Te units above approximately $x = 10$ % (roughly the composition at which $m(\langle n \rangle)$ begins to deviate).

Taken together with the over-abundance of Te - O - Te connections and under abundance of P - O - Te connections above $x = 15$ mol % (see

Table 8), we propose a model in which some fraction, $g_{\text{ES}}(x)$, of the Te atoms form edge-sharing connections with another Te atom. Applying the same topological corrections described previously for the single redundant BO bond involved, we obtain:

$$\langle \phi \rangle = 2f_p + f_{\text{Te}}(3g_{\text{ES}} + 4(1 - g_{\text{ES}})), \quad (15)$$

in which the only unknown is the compositional function for $g_{\text{ES}}(x)$. To deal with this unknown, we employ an approach similar to that used to treat small rings present in alkali germanate glasses [15] where the functional form for the fraction of rings was guided by the experimentally observed intensity of the corresponding Raman breathing mode. Since the 615 cm^{-1} mode first appears near 10 mol %, increases rapidly up to around 25 mol % and then appears to saturate [33], we propose a three-parameter model using a simple "rising" exponential of the form:

$$g_{\text{ES}}(x) = g_{\infty}[1 - \exp(-(x - x_0)/B)], \quad (16)$$

which begins rising at x_0 to a maximum fraction g_{∞} at large x . In applying the model, we have adjusted the three parameters in a manner consistent with the intensity of the Raman mode and obtained a reasonable match ($\chi_{\text{red}}^2 = 1.0$) of $m(\langle \phi \rangle)$ with the canonical curve (see Fig. 10) for $g_{\infty} = 0.8$, $x_0 = 6$ and $B = 10$.

5. Discussion

The fundamental feature of the MNF effect is the deviation of properties from simple linear mixing. Others have emphasized the non-linear trends for ion conductivity as a well-known example of the MNF effect. Is there such an effect for either the glass transition temperature or the fragility that we have measured in these five MNF systems? The short answer is yes, as both these properties deviate from the lines connecting them between NaPO_3 and the other NFO. For T_g these deviations were indicated in Figs. 5, 6 and 10 whenever the T_g of the NFO was known. As for the fragility, we plot in Fig. 11 the collective values for all five systems as a function of the added NFO. The fragility of the NFO endmember is known only for GeO_2 ($m = 17$) and B_2O_3 ($m \approx 30$) and for these two the fragility clearly deviates negatively from the linear prediction based on simple mixing.

In the absence of any linear dependence on molar mixing ratio, we consider other possible parameters. For the glass transition temperature we would suggest that T_g most closely tracks the BO bond density $\langle n \rangle$. This is evident in Fig. 12 where (with the possible exception of Te) we see that T_g increases with $\langle n \rangle$ in a linear manner (with correlation coefficients, $R^2 > 0.90$). As for the fragility, we would instead argue that the best coordinate is $\langle \phi \rangle$, the average bond density topologically

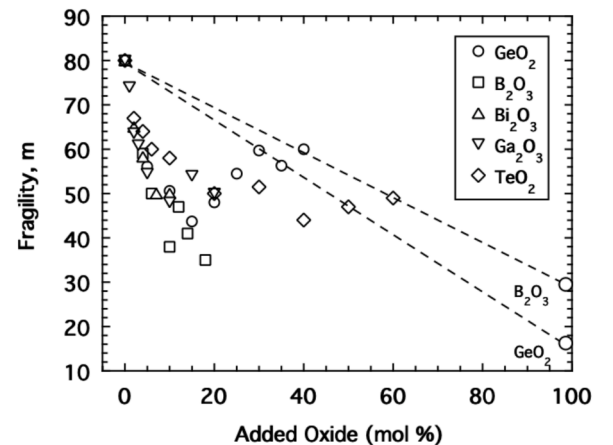


Fig. 11. The fragility index of $(\text{NFO})_x(\text{NaPO}_3)_{100-x}$ plotted against added NFO (see key). Dashed lines indicate predictions for B_2O_3 and GeO_2 if linear mixing is applied.

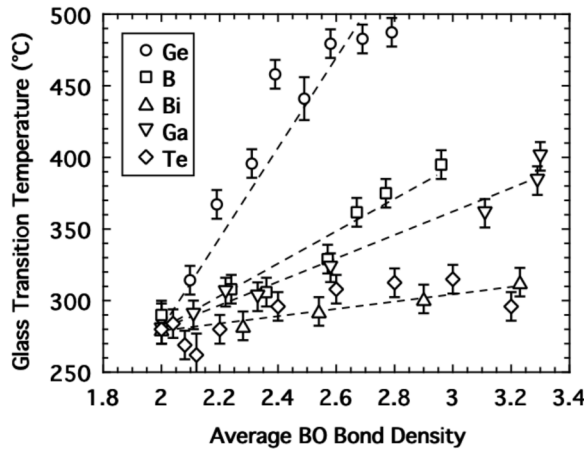


Fig. 12. The glass transition temperature plotted against average bridging oxygen bond density for all five MNFs studied. Dashed lines are linear fit (fixed to $T_g = 280$ °C of NaPO_3 at $\langle n \rangle = 2$) with correlation coefficients: $R^2 = 0.86$ (Ge), 0.93 (B), 0.96 (Bi), 0.97 (Ga) and 0.57 (Te).

corrected for SSUs. In all five system we could identify possible SSUs and, by applying a now well-developed procedure for topologically correcting for redundant BO bonds, have managed to collapse $m(\langle \phi \rangle)$ onto the canonical curve. We acknowledge that this procedure is weakest for the case of Ga_2O_3 where we resorted to an ad hoc hypothesis that edge-sharing occurs in a like fashion to that found in Bi_2O_3 . On the other hand, our modeling of SSUs for GeO_2 , B_2O_3 and TeO_2 are all fully consistent with how Eckert [26] ranked these three alloys with regards to deviations from a purely random network formation. Of the three, Eckert noted that GeO_2 exhibited bond distributions most nearly matching a random connectivity while B_2O_3 exhibited an overabundance of heteroatomic (B - O - P) connections [10] (consistent with the two structures illustrated in Fig. 7) and TeO_2 an overabundance of homoatomic (Te - O - Te) connections [33] (consistent with the edge-sharing Te illustrated in Fig. 10).

Over the years since we first proposed the canonical curve and the topological corrections involving SSUs, we have encountered two main criticisms. Firstly, that the canonical curve itself lacks *validity* as it is purely empirical and largely determined by just two network-forming glasses for which SSUs are presumed to be absent. And secondly, that the procedure by which we correct for the topology of SSUs amounts to a form of circular logic in which an *arbitrary* manipulation of our data is being performed with the predestined intent of collapsing the fragility onto the canonical curve. As to the validity of the canonical curve itself, we acknowledge that it was largely developed from empirical observations and has received only limited theoretical framing [11]. It is defined largely by two network forming glasses, one a series of oxide-free chalcogenide glasses ($\text{As}_x\text{Ge}_{100-2x}$) described by an atomic bond density $\langle r \rangle$ and the other a series of sodium phosphate glasses described by a bridging oxygen bond density ranging from $\langle n \rangle = 2$ to 3. To date there has been no evidence for the formation of small rings nor edge-sharing in either glass series. Furthermore, the fact that these two glass systems collapse *only* when $\langle r \rangle$ is used to represent the chalcogenide network of atomic linkages and $\langle n \rangle$ is used to represent the phosphate network of tetrahedra interconnected by bridging oxygen, in and of itself demonstrates that fragility must reflect the topological connectivity of these semi-flexible networks.

As to the criticism that the procedure by which we correct for the topology of SSUs is arbitrary data manipulation, we emphasize the high degree of complexity in constructing the data transformations given by Eqs. (7), 9, 11, 13 and 15. In fact, one can show that these data transformations (and others we have used in the past) have a general form:

$$\langle \phi \rangle = \langle n \rangle - g(x)\{F_1(x)\} - g^2(x)\{F_2(x)\}, \quad (17)$$

where the $F_{1,2}(x)$ are functional *constraints* specific to each situation which in turn involve a rather complex convolution of the glass chemistry, the NMR bond distributions and the particular type of SSU being topologically corrected. The only adjustable feature is the function, $g(x)$, chosen to model the evolution of the SSUs whose functional form and parameters are guided by other experimental evidence (usually Raman intensities). The resulting data transformation in Eq. (17) therefore involves a complex weaving together of topological corrections, compositional constraints and possibly one adjustable function. It has very few degrees of freedom and is in no way an arbitrary manipulation of the data. Furthermore, inclusion of the adjustable function $g(x)$ poses no more circular logic than that of applying a conventional curve fit to a set of fixed data (e.g., the stretched exponentials fit to autocorrelation functions in Fig. 1). Instead of fitting a curve to a fixed set of data, we are merely fitting data to a fixed curve and applying the same statistical evaluation of the goodness of fit.

Lastly, a few comments about the secondary relaxation. This relaxation which appears some 5 orders of magnitude more slowly than the α -relaxation, was not a focus of the current investigation. However, this slow diffusive mode appears to be scattering from the cations of the added NFO which become incorporated into the network. Despite their much slower timescale (in the wavevector regime of our light scattering experiment), these slow relaxations appear to be related to the α -relaxation as they closely match it when viewed at a wavevector range near $q \sim 1 \text{ \AA}^{-1}$ for which the RMS displacement has only progressed a few Angstroms. To the extent that one can use $\tau_s(q \sim 1 \text{ \AA}^{-1})$ as a surrogate for the α -relaxation time, we can in principle extend measurements of τ_α in PCS to cover an impressive 12 decades (see e.g., Fig. 4) using a single experimental technique. This second relaxation is consistently more exponential than the α -relaxation with β_s ranging from $\beta_s \approx 0.6$ to $\beta_s \approx 0.8$. Moreover, the relative strength of this secondary relaxation (in comparison with the α -relaxation) is largest for TeO_2 and Bi_2O_3 where depolarized scattering was needed to separate out the α -relaxation. This is fully consistent with their greater atomic form factors, each of which scale as the number electrons ($Z = 52$ and 83, respectively), compounded also by the fact that scattering intensity depends on the *square* of the form factor [37].

6. Conclusion

The mixed network former effect has received much attention for solid state properties but little examination for properties in the liquid state. Here we have used dynamic light scattering to probe the primary viscous relaxation mode in the dynamic structure factor and assess how both the glass transition temperature and the fragility are influenced by the mixing of a second network-forming oxide with sodium metaphosphates. Generally, the addition of the second NFO results in increased BO bond density due to crosslinking of phosphate chains which is seen to directly increase the glass transition temperature. The fragility, however, exhibits contradictory trends in which it usually decreases (but sometimes increases) with increasing BO bond density. Such dissonance was encountered previously in binary alkali oxides where a topological correction to the BO bond density afforded a better understanding of how fragility depends on network structure. The topological correction arises from the presence of small, crystallite-like super-structural units that possess redundant BO connections. Although not the main objective of the present study, we are able to identify super-structural units in many of the MNF reported here for which the topological correction results in a single, common dependence of fragility on the bond density of the network.

CRedit authorship contribution statement

David L. Sidebottom: Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding

acquisition, Conceptualization. **Jay McCown:** Investigation, Formal analysis, Data curation. **Josh Kevil:** Investigation, Formal analysis, Data curation. **Daniel White:** Investigation, Formal analysis, Data curation. **Daniel Olabode:** Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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